PhantomChannel: Artifact Appendix

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A ARTIFACT APPENDIX

A.1 Abstract

PhantomChannel is a research prototype for embedding auxiliary symbols beyond BLE packet parsing boundaries and recovering them with an SDR receiver. The artifact provides transmitter and receiver source, firmware images, software tests, demonstration videos, and an offline experiment bundle. This appendix outlines access, requirements, and the evaluation workflow; the linked documentation provides detailed commands and configuration notes. Project-owned source is MIT licensed, and videos are CC BY 4.0. Third-party components retain their file-level and vendor terms, as documented in `THIRD_PARTY_NOTICES.md`.

A.2 Description & Requirements

The source bundle contains the receiver, platform-specific firmware, controller patches, configuration examples, tools, and tests. The experiment bundle contains analysis-ready records, analysis scripts, regression references, and a top-level runner. Each bundle includes a README; `DEPENDENCY_MAP.md` relates experiment inputs to generated results. The online artifact guide is the maintained entry point for setup, supported workflows, output locations, and implementation notes.

The default evaluation path replays retained records offline. Live hardware experiments are optional and use the separate platform guides. The appendix covers the available replay paths; the artifact guide distinguishes recomputed results, archived summaries, and analyses requiring additional experimental evidence.

A.2.1 Security, privacy, and ethical concerns. Offline replay requires no privileged access or radio transmission. Hardware experiments replace development-board firmware and emit BLE traffic; use owned devices and authorized, isolated RF setups under applicable spectrum rules. Published records use anonymous groups and relative timing. Keep credentials, device identifiers, and unrelated traffic out of shared configurations and captures.

A.2.2 How to access. The archived artifact is available at doi:10.5281/zenodo.22842761. Download the source and experiment bundles together with the record's README and checksum manifest. The maintained repository is https://github.com/Tincyu/PhantomChannel. Start with the online artifact guide for current documentation and the links to supplemental environment files.

The DOI identifies the deposited snapshot; the linked repository documentation may evolve with platform support and maintenance updates. To repeat a run, retain the archive checksums, the repository commit used for supplemental files, and the installed dependency manifest. Updated analyses should identify their code and data versions separately from the archived replay.

A.2.3 Hardware dependencies. Offline evaluation runs on a conventional CPU workstation with enough memory and disk space for bundle extraction and generated outputs. No SDR, GPU, or BLE development board is needed for this path. Live evaluation requires a supported transmitter, a compatible BLE peer where applicable, and an SDR receiver. The firmware guide provides the board/image matrix and programming instructions; receiver setup and supported configurations are linked from the artifact guide.

A.2.4 Software dependencies. The offline workflow uses Linux, Python, virtual environments, and standard scientific Python packages, including NumPy, SciPy, and Matplotlib. Source checks additionally use the test dependencies supplied with the source bundle. The environment guide records tested Python/OS combinations, exact dependency pins, and commands for capturing the installed environment. Use its versioned requirements for the selected workflow rather than assuming that arbitrary package upgrades are equivalent. Live reception and firmware builds require additional receiver runtimes, vendor SDKs, and toolchains documented in the platform guides; vendor SDKs are not bundled.

A.2.5 Benchmarks. The experiment bundle supplies IQ packet windows, parser candidates, advertising records, connected timing vectors, bandwidth and Data-ablation records, detector features and scores, and auxiliary NLOS and antenna records. These are experiment-specific workloads. The Data portion of the ablation table is covered by the supplied replay. Real-boundary detector tables are archived diagnostics; independent reconstruction of their packet scores is outside this appendix's replay scope. Input provenance and result coverage are described in the artifact guide and dependency map.

A.3 Set Up

A.3.1 Installation. Download and extract the bundles into separate directories. Verify the available downloads against the record's manifest:

```
sha256sum --ignore-missing -c SHA256SUMS
```

Follow the environment guide to select the tested Python environment and install the appropriate pinned dependencies. Create separate virtual environments for source tests and experiment replay. For example, in the source root:

```
python3 -m venv .venv
.venv/bin/python -m pip install \
  -r requirements-test.txt
```

The environment guide supplies the corresponding experiment installation commands and dependency files. Installation requires package downloads or a compatible wheel cache; subsequent archived replay is offline. Save the OS, Python, dependency, and input-version information with each run.

A.3.2 Basic test. In the prepared source environment, run:

```
.venv/bin/python -m pytest -q tests
```

Expect successful completion without failed tests. In the prepared experiment root, the table runner provides a basic input/analysis check:

```
.venv/bin/python scripts/recompute_tables.py
```

Successful completion creates CSVs under `outputs/recomputed_tables/`. The artifact guide explains how these checks relate to the full replay.

A.4 Evaluation Workflow

A.4.1 Major claims. The artifact supports the following aspects of the paper's evaluation within the replay scope described above:

- (C1) Receiver behavior across recorded bandwidth, range, SNR, and BPL/CAF conditions (Fig. 10 and Tables 3–4; E1).
- (C2) Protocol-visible timing for the retained advertising and connected packet groups (Fig. 12 and Table 5; E2).
- (C3) Spectral and tail-energy observability using retained IQ windows, eligible features, and supported session-held-out score views (Fig. 11 and the countermeasure evaluation; E3).

A.4.2 Experiments. After installation, run the archived workflow from the experiment root:

```
.venv/bin/python reproduce_all.py
```

Expect successful completion, `[OK]` checks, plots, and CSV/JSON results. The online artifact guide links the experiment-specific commands, input-to-output map, and interpretation notes. Regression checks establish agreement with the archived analysis baseline; they do not independently validate every historical paper label. As a planning estimate, setup takes about half an hour, while each experiment group requires a few person-minutes for inspection and several compute-minutes on a conventional workstation. Downloads and host performance can change elapsed time.

- (E1) **Receiver and auxiliary replay (C1).** *Preparation:* use the supplied range, bandwidth, Data-ablation, and auxiliary records with their documented selection rules. *Execution:* run the full replay and follow the receiver/auxiliary section of the artifact guide for focused inspection. *Results:* inspect the overhead, range, and SNR plots and the table and auxiliary summaries under `outputs/`. Interpret candidate recovery, full-frame rate, and correctly recovered useful payload separately. The guide specifies counting rules, integrity requirements, framing overhead, time denominators, and the optional correctness-aware antenna replay.
- (E2) **Timing replay (C2).** *Preparation:* preserve packet grouping, timing vectors, and sample masks. *Execution:* run the full replay or the table runner above. *Results:* inspect the advertising timing plots and the connected timing CSV. Trace conditions through `Input_dataset`; dataset aliases alone do not establish transmitted payload length. Interpret summaries for the retained samples under the documented filtering rules. The guide records provenance and any distinctions from historical paper summaries.
- (E3) **Spectral and detector replay (C3).** *Preparation:* preserve IQ group identities, feature eligibility, and session grouping. *Execution:* run the full replay; the guide also lists individual analysis entry points. *Results:* inspect the frequency-similarity CSV, PIP pairwise matrix, and tail-energy outputs. PIP spectral evidence is the cross-group statistic, not a single illustrative overlay. Distinguish feature-based and session-held-out replay from archived real-boundary summaries. Consult the guide for the applicable input mapping, parameter context, and supported detector views.

A.5 Notes on Reusability

The linked guides maintain detailed configuration, compatibility notes, and workflow updates without requiring this appendix to enumerate every platform or package version. Preserve schemas, grouping, sample masks, and metric definitions when replacing inputs. Record changed code, data, or processing rules as a new analysis, and retain its outputs and environment manifest. Hardware/RF acceptance requires a separate live run following the platform guides. Questions and replication reports can be submitted through the project issue tracker.

A.6 Version

Based on the LaTeX template for Artifact Evaluation V20220926.